

**GRADE 10 — SENIOR SCHOOL (CBC)**

# **PHYSICS**

Learner's Notes & Activity Book

## **LEARNER NOTES**

Covering: Mechanics & Thermal Physics • Waves & Optics •  
Electricity & Magnetism • Environmental & Space Physics

Based on the KICD Grade 10 Senior School Physics Curriculum Design (2024)

Prepared for learners aged 14–16 years

Contains: Notes, Idea Boxes, Case Studies, Reflection Questions, Worksheets, Exercises and an Answer Key

# Table of Contents

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## How to Use These Notes

### STRAND 1: MECHANICS AND THERMAL PHYSICS

- 1 Introduction to Physics
- 2 Pressure
- 3 Mechanical Properties of Materials
- 4 Temperature and Thermal Expansion
- 5 Moments and Equilibrium
- 6 Energy, Work, Power and Machines

### STRAND 2: WAVES AND OPTICS

- 7 Properties of Waves
- 8 Radioactivity and Stability of Isotopes

### STRAND 3: ELECTRICITY AND MAGNETISM

- 9 Electrostatics
- 10 Current Electricity
- 11 Introduction to Electronics

### STRAND 4: ENVIRONMENTAL AND SPACE PHYSICS

- 12 Greenhouse Effect and Climate Change
- 13 Introduction to Space Physics

### ANSWER KEY

Answers to All Worksheets, Exercises and End of Topic Questions

# How to Use These Notes

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Welcome to your Grade 10 Physics Learner Notes! This book has been written especially for you, following the Kenya Institute of Curriculum Development (KICD) Senior School Physics Curriculum Design. As you go through each topic, look out for these boxes:

 **Idea Box** – an important fact, tip, or shortcut to remember.

 **Case Study** – a real-life Kenyan or global example showing how the topic is applied.

 **Reflection Question** – a question with no single "correct" answer; think about it and discuss with classmates.

 **Worksheet/Exercise** – practice questions to attempt on your own or in groups.

 **End of Topic Questions** – a short revision test at the end of every topic.

All answers to worksheets, exercises and end of topic questions are found in the **Answer Key** chapter at the back of this book. Try to attempt every question first before checking the answers!

# STRAND 1: MECHANICS AND THERMAL PHYSICS

Topics 1–6: Introduction to Physics, Pressure, Mechanical Properties of Materials, Temperature and Thermal Expansion, Moments and Equilibrium, Energy Work Power and Machines

# Topic 1: Introduction to Physics

Strand 1.1 • Suggested lessons: 6

## By the end of this topic, you should be able to:

- Explain Physics as a body of knowledge in science
- Describe the branches of Physics as a field of study
- Outline the importance of Physics in day-to-day life
- Relate Physics to other fields of study
- Identify possible career opportunities in the field of Physics
- Appreciate the importance of Physics in day-to-day life

*Key Inquiry Question: How is Physics relevant in day-to-day life?*

## 1.1 What is Physics?

**Physics** is a branch of science that studies matter, energy, and the way they interact with each other. It seeks to explain natural phenomena – from why a mango falls from a tree, to how a radio sends signals, to why the sky appears blue. Physics uses observation, measurement and experiments to develop laws and theories that describe how the universe behaves.

As a body of knowledge, Physics is built on concepts that have been tested repeatedly through experiments. A **law** in Physics is a statement, often expressed mathematically, that describes a pattern observed in nature under certain conditions (e.g. Hooke's Law). A **theory** is a well-tested explanation of why that pattern occurs.

## 1.2 Branches of Physics

| Branch                  | What it studies                                                 |
|-------------------------|-----------------------------------------------------------------|
| Mechanics               | Motion of objects, forces, energy and machines                  |
| Thermodynamics          | Heat, temperature and energy transfer                           |
| Electricity & Magnetism | Electric charges, currents and magnetic effects                 |
| Waves                   | Sound, water waves, and other vibrations that transfer energy   |
| Geometrical Optics      | Behaviour of light: reflection, refraction, lenses and mirrors  |
| Electronics             | Behaviour of semiconductor devices used in circuits and gadgets |
| Modern Physics          | Atomic and nuclear physics, radioactivity, relativity           |
| Astronomy/Astrophysics  | Stars, planets, galaxies and the universe                       |

### Remember

A simple way to remember the branches of Physics is the acronym **M.E.T.W.O.E.M.A**: Mechanics, Electricity & Magnetism, Thermodynamics, Waves, Optics, Electronics, Modern Physics, Astronomy.

## 1.3 Importance of Physics in Day-to-Day Life

- Helps us understand and explain natural events (rain formation, lightning, echoes)
- Basis for technology: mobile phones, solar panels, motor vehicles, medical scanners
- Improves safety: understanding forces helps engineers design safer bridges and buildings
- Supports other sciences such as Chemistry, Biology, Geography and Agriculture
- Helps in energy production and conservation (electricity generation, biogas, solar energy)

## 1.4 Physics and Other Fields of Study

Physics is closely linked to many other subjects:

- **Mathematics** – provides the language (equations, graphs) used to describe physical laws
- **Chemistry** – atomic structure and bonding rely on physical principles
- **Biology** – understanding of forces, levers (bones and muscles), and optics (the eye)

- **Geography** – atmospheric pressure, climate and weather patterns
- **Agriculture** – irrigation pumps, greenhouse technology, machinery

### **Case Study: Physics Behind the Jua Kali Sector**

Kenya's Jua Kali artisans design and fabricate wheelbarrows, ox-ploughs, water pumps, and metal gates. Every one of these products relies on physics: the wheelbarrow uses the principle of moments (Topic 5) to make lifting loads easier; welded metal gates must resist bending forces (mechanical properties of materials, Topic 3); water pumps rely on pressure differences (Topic 2). This shows how physics knowledge, even without a university degree, drives innovation and self-employment in Kenya.

## **1.5 Careers in Physics**

Studying Physics opens up many career pathways beyond Senior School, including:

- Engineering (electrical, mechanical, civil, aeronautical, telecommunications)
- Medicine and medical imaging (radiography, radiotherapy)
- Meteorology and climate science
- Astronomy and space science
- Renewable energy technology (solar, wind, geothermal)
- Aviation and pilot training
- Research and university lecturing
- Information and communication technology

### **Reflection**

Physics is sometimes seen as a "difficult" or "boys only" subject. Why do you think this myth exists, and what can you and your classmates do to encourage everyone – regardless of gender – to enjoy and excel in Physics?

### **Key Terms**

Physics • Law • Theory • Mechanics • Thermodynamics • Astrophysics • Branch of science

### **Worksheet 1.1**

1. In groups of four, list five gadgets you used today. For each, name at least one branch of Physics that explains how it works.
2. Design a simple career chart (on manila paper) showing three careers related to Physics and the Physics knowledge needed for each.
3. Interview a local artisan, technician, or engineer and find out how they use Physics knowledge in their daily work. Present your findings to the class.

### **End of Topic 1 Questions**

1. Define the term "Physics".
2. State any four branches of Physics.
3. Explain the difference between a scientific law and a scientific theory.
4. Give three ways in which Physics is applied in agriculture.
5. Name three careers a learner who studies Physics can pursue after Senior School.
6. Explain, using an example, how Physics is related to Biology.

## Topic 2: Pressure

Strand 1.2 • Suggested lessons: 25

### By the end of this topic, you should be able to:

- Describe atmospheric pressure as used in Physics
- Demonstrate the existence of atmospheric pressure in nature
- Investigate factors affecting pressure in fluids
- Apply the equation  $P = \rho gh$  to determine pressure in fluids
- Demonstrate transmission of pressure in fluids
- Appreciate applications of pressure in day-to-day life

*Key Inquiry Question: How does density of fluid, acceleration due to gravity and depth below the free surface affect pressure in a fluid?*

### 2.1 Meaning of Pressure

**Pressure** is defined as the force acting normally (perpendicularly) per unit area.

$$\text{Pressure (P)} = \text{Force (F)} \div \text{Area (A)}$$

The SI unit of pressure is the **pascal (Pa)**, where  $1 \text{ Pa} = 1 \text{ N/m}^2$ . Other units include  $\text{N/cm}^2$ , atmospheres (atm), millibars, and millimetres of mercury (mmHg).

#### Idea Box

For the same force, a smaller area produces greater pressure. This is why a sharp knife (small area) cuts more easily than a blunt one, and why a nail has a pointed tip.

### 2.2 Atmospheric Pressure

**Atmospheric pressure** is the pressure exerted on the earth's surface (and everything on it) by the weight of air in the atmosphere above. At sea level, standard atmospheric pressure is about  $101,325 \text{ Pa}$  (1 atm). Atmospheric pressure decreases with increasing altitude because there is less air above you.

Evidence for atmospheric pressure includes: the collapsing can experiment (a sealed tin collapses when the air inside is heated and then cooled rapidly), the Magdeburg hemispheres experiment, and the simple observation that a drinking straw allows you to suck up a liquid because you reduce the pressure inside the straw below atmospheric pressure.

### 2.3 Pressure in Fluids (Liquids)

Pressure in a liquid at a given depth depends on three factors:

1. Depth (h) below the free surface – the deeper you go, the higher the pressure
2. Density ( $\rho$ ) of the liquid – denser liquids exert greater pressure at the same depth
3. Acceleration due to gravity (g)

These are combined in the equation:

$$P = \rho gh$$

where P is pressure (Pa),  $\rho$  is density ( $\text{kg/m}^3$ ), g is acceleration due to gravity (approximately  $10 \text{ m/s}^2$  or  $9.8 \text{ m/s}^2$ ), and h is the depth (m).

#### Worked Example

Find the pressure exerted by water at a depth of 5 m in a dam. (Density of water =  $1000 \text{ kg/m}^3$ ,  $g = 10 \text{ m/s}^2$ )

$$P = \rho gh = 1000 \times 10 \times 5 = \mathbf{50,000 \text{ Pa (50 kPa)}}$$

Important properties of fluid pressure:

- Pressure in a fluid at rest acts equally in all directions at a given point
- Pressure increases with depth
- Pressure at a point does not depend on the shape or cross-sectional area of the container, only on the depth and density of the fluid

## 2.4 Transmission of Pressure in Fluids - Pascal's Principle

**Pascal's Principle** states that pressure applied to an enclosed fluid is transmitted equally to all parts of the fluid and to the walls of the container. This principle is applied in hydraulic machines such as the hydraulic jack, hydraulic press and hydraulic brakes used in motor vehicles.

 **Idea Box**

In a hydraulic machine, a small force applied on a small piston creates the same pressure that acts on a much larger piston, producing a much larger force. This is why hydraulic jacks can lift heavy vehicles with little effort!

## 2.5 Applications of Pressure in Fluids

| Device             | How it applies pressure principles                                                                |
|--------------------|---------------------------------------------------------------------------------------------------|
| Drinking straw     | Sucking reduces pressure inside the straw; atmospheric pressure pushes the liquid up              |
| Syringe            | Pulling the plunger reduces pressure, drawing in liquid; pushing increases pressure, expelling it |
| Siphon             | Uses atmospheric pressure and gravity to move liquid over a raised section of tube                |
| Hydraulic machines | Use Pascal's principle to multiply force (jacks, car brakes, presses)                             |
| Bicycle pump       | Compresses air, increasing its pressure, to force it into the tyre                                |
| Water pumps        | Create a pressure difference to move water from a low point (e.g. a well) to a tank               |

 **Case Study: Water Pumping in Rural Kenya**

Many rural communities in Kenya rely on hand pumps and motorised pumps to draw water from boreholes. These pumps work by creating a region of low pressure that allows atmospheric pressure (or mechanical force) to push water up the pipe. Understanding the limits of suction pumps (they typically cannot lift water more than about 10 metres using atmospheric pressure alone) has guided engineers to design submersible pumps for deep boreholes, which push rather than suck water upward.

 **Reflection**

Why do mountaineers climbing Mount Kenya or Mount Kilimanjaro sometimes experience difficulty breathing, and why do their cooking times for food increase at high altitude?

**Key Terms**

Pressure • Pascal (Pa) • Atmospheric pressure • Density • Pascal's Principle • Hydraulic machine

 **Worksheet 2.1**

1. A rectangular block of mass 20 kg rests on a table. Its base measures 0.5 m by 0.4 m. Calculate the pressure it exerts on the table. (Take  $g = 10 \text{ m/s}^2$ )
2. Calculate the pressure exerted by a column of mercury of height 0.76 m. (Density of mercury =  $13,600 \text{ kg/m}^3$ ,  $g = 10 \text{ m/s}^2$ )
3. Explain why dams are built thicker at the bottom than at the top.
4. Using a labelled diagram, explain how a bicycle pump works.
5. In groups, design a simple experiment using a plastic bottle and water to demonstrate that pressure increases with depth.

1. Define pressure and state its SI unit.
2. State the three factors that affect pressure in a liquid.
3. Write the equation used to calculate pressure in a fluid, defining each symbol.
4. State Pascal's Principle.
5. A diver is swimming at a depth of 10 m in sea water of density  $1030 \text{ kg/m}^3$ . Calculate the pressure due to the water at this depth. ( $g = 10 \text{ m/s}^2$ )
6. Explain how a hydraulic jack is able to lift a heavy car using a small applied force.
7. Give two everyday applications of atmospheric pressure.

# Topic 3: Mechanical Properties of Materials

Strand 1.3 • Suggested lessons: 10

**By the end of this topic, you should be able to:**

- Explain the mechanical properties of materials
- Demonstrate the mechanical properties of materials
- Determine tensile stress and strain using mathematical formulae
- Describe applications of mechanical properties of materials
- Appreciate the importance of these properties in day-to-day life

*Key Inquiry Questions: (1) Why does a string snap easily compared to a spring? (2) Why is it important to study the mechanical properties of materials?*

## 3.1 Mechanical Properties of Materials

| Property     | Meaning                                                                                             |
|--------------|-----------------------------------------------------------------------------------------------------|
| Elasticity   | Ability of a material to return to its original shape/size after a deforming force is removed       |
| Plasticity   | Ability of a material to retain a new shape after a deforming force is removed                      |
| Ductility    | Ability of a material to be drawn/stretched into a thin wire without breaking                       |
| Malleability | Ability of a material to be hammered or rolled into thin sheets without breaking                    |
| Brittleness  | Tendency of a material to break/shatter without much deformation (little or no plastic deformation) |
| Strength     | Ability of a material to withstand an applied force without breaking                                |
| Hardness     | Resistance of a material to scratching, cutting, or indentation                                     |
| Stiffness    | Resistance of a material to deformation (bending or stretching)                                     |

 **Idea Box**

Glass and chalk are brittle – they shatter with little warning. Copper and gold are ductile and malleable, which is why copper is drawn into electrical wires and gold is hammered into thin jewellery.

## 3.2 Hooke's Law

When a spring (or wire) is stretched by a force within its elastic limit, the extension produced is directly proportional to the applied force. This is **Hooke's Law**:

$$F = k\Delta L$$

where F is the applied force (N), ΔL is the extension (m), and k is the spring constant / constant of elasticity (N/m) – a measure of the stiffness of the material.

Beyond a certain force (the **elastic limit**), the material no longer obeys Hooke's Law and will not return to its original length when the force is removed – it becomes permanently deformed.

## 3.3 Stress, Strain and Young's Modulus

$$\text{Stress} = F \div A$$

Tensile **stress** is the force applied per unit cross-sectional area of a material. Its SI unit is the pascal (Pa), same as pressure.

$$\text{Strain} = \Delta L \div L_0$$

Tensile **strain** is the ratio of extension to original length. Strain has no units (it is a ratio).

$$Y = \text{Stress} \div \text{Strain}$$

**Young's Modulus (Y)** is the ratio of stress to strain for a material within its elastic limit. It measures the stiffness of a material and is measured in pascals (Pa).

#### Worked Example

A wire of original length 2.0 m and cross-sectional area  $1.0 \times 10^{-6} \text{ m}^2$  extends by 1.0 mm when a force of 40 N is applied. Calculate: (a) the stress, (b) the strain, (c) Young's Modulus.

(a)  $\text{Stress} = F/A = 40 / (1.0 \times 10^{-6}) = 4.0 \times 10^7 \text{ Pa}$

(b)  $\text{Strain} = \Delta L/L_0 = 0.001 / 2.0 = 5.0 \times 10^{-4}$

(c)  $Y = \text{Stress}/\text{Strain} = (4.0 \times 10^7) / (5.0 \times 10^{-4}) = \mathbf{8.0 \times 10^{10} \text{ Pa}}$

### 3.4 Industrial Applications of Mechanical Properties

- **Construction:** steel reinforcement bars (high tensile strength) are used in concrete to resist tension, since concrete alone is strong in compression but weak in tension
- **Cables and wires:** ductile metals like copper and aluminium are drawn into electrical cables
- **Vehicle bodies:** materials with the right balance of strength, stiffness and ability to absorb impact (e.g. some plastics and alloys) are chosen for safety
- **Springs** in vehicle suspension and mattresses rely on elasticity
- **Cutting tools** require hardness (e.g. tungsten carbide, diamond-tipped blades)

#### Case Study: Choosing Materials for the SGR Bridges

The Standard Gauge Railway (SGR) in Kenya crosses valleys using long steel-and-concrete bridges. Engineers select steel for its high tensile strength and elasticity (it can bend slightly under load and return to shape), while concrete is used for its compressive strength. Understanding mechanical properties allows engineers to combine materials so that each handles the type of force it is best suited for – this is called composite design.

#### Reflection

Why do you think a rubber band can stretch to several times its length and return to normal, while a metal wire snaps after stretching only slightly? What does this tell you about the elastic limits of different materials?

#### Key Terms

Elasticity • Plasticity • Ductility • Malleability • Brittleness • Hardness • Stiffness • Hooke's Law • Stress • Strain • Young's Modulus • Elastic limit

#### Worksheet 3.1

1. A spring of natural length 15 cm extends to 18 cm when a force of 6 N is applied. Calculate the spring constant.
2. A metal wire of cross-sectional area  $2.0 \times 10^{-6} \text{ m}^2$  is stretched by a force of 100 N. Calculate the stress in the wire.
3. Classify the following materials as ductile, malleable, brittle, or elastic: copper wire, glass, rubber band, gold leaf, chalk.
4. In pairs, stretch a rubber band and a thin thread with similar loads. Record and compare their behaviour. Which one obeys Hooke's Law over a larger range?

#### End of Topic 3 Questions

1. Define the terms ductility, malleability and brittleness.
2. State Hooke's Law.
3. Write the formulae for tensile stress, tensile strain and Young's Modulus, defining each symbol.
4. A wire extends by 0.5 mm from an original length of 1.0 m under a certain load. Calculate the strain.

5. Explain why steel is preferred over glass for constructing bridge cables.
6. Distinguish between elasticity and plasticity.

# Topic 4: Temperature and Thermal Expansion

Strand 1.4 • Suggested lessons: 10

**By the end of this topic, you should be able to:**

- Explain the meaning of temperature as used in thermal physics
- Measure temperature using different technologies
- Investigate thermal expansion and contraction in solids and fluids
- Describe applications of thermal expansion in solids and fluids
- Appreciate the applications of thermal expansion in day-to-day life

Key Inquiry Questions: (1) Why is the lid of a sufuria made wider? (2) Why does a glass bottle break when the water in it freezes?

## 4.1 Meaning of Temperature

**Temperature** is a measure of the degree of hotness or coldness of a body, or more precisely, a measure of the average kinetic energy of the particles making up a substance. Temperature is measured in degrees Celsius (°C) or in Kelvin (K), the SI unit. To convert Celsius to Kelvin, add 273:

$$T(K) = T(^{\circ}C) + 273$$

## 4.2 Measuring Temperature

| Device                                             | Principle used                                                                     |
|----------------------------------------------------|------------------------------------------------------------------------------------|
| Liquid-in-glass thermometer                        | Expansion of a liquid (mercury/alcohol) in a narrow tube                           |
| Bimetallic strip / thermostat                      | Two different metals expand at different rates, causing bending                    |
| Thermocouple                                       | Voltage generated at the junction of two different metals changes with temperature |
| Resistance Temperature Detector (RTD) / Thermistor | Electrical resistance of a material changes with temperature                       |
| Infrared thermometer                               | Detects infrared radiation emitted by an object (no contact needed)                |
| Silicon diode sensor                               | Voltage across a diode changes predictably with temperature                        |

 **Idea Box**

Digital medical thermometers commonly use thermistors, while infrared "gun" thermometers (widely used during the COVID-19 pandemic) measure body temperature without touching the skin.

## 4.3 Thermal Expansion in Solids

When a solid is heated, its particles vibrate more and move slightly further apart, causing the solid to expand. Different materials expand by different amounts for the same temperature rise – this is described by the **linear expansivity** of a material, which is the increase in length per unit original length per degree rise in temperature.

 **Idea Box**

Metals like aluminium expand more than steel or copper for the same temperature change. This is why railway lines and bridges are built with small expansion gaps to prevent buckling on hot days.

## 4.4 Thermal Expansion in Liquids - the Unusual Expansion of Water

Most liquids expand uniformly on heating. Water, however, behaves unusually: as it is cooled from 4°C to 0°C, it actually **expands** instead of contracting, and reaches its maximum density at 4°C. This explains why ice floats on water, and why a sealed glass bottle full of water can crack when the water inside freezes and expands.

### Case Study: Fish Survival in Cold Lakes

The unusual expansion of water is vital for aquatic life. When a lake's surface water cools below 4°C in cold climates, the water becomes less dense and stays on top, eventually freezing into a layer of ice. This ice layer insulates the water below, which remains at about 4°C, allowing fish and other organisms to survive underneath even when the surface is frozen solid.

## 4.5 Applications of Thermal Expansion

| Application                                   | How it works                                                                                     |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------|
| Thermostat (bimetallic strip)                 | Bends when heated, breaking or completing an electrical circuit to control heaters, irons, ovens |
| Fire alarm / flash indicator                  | Bimetallic strip bends under heat from fire, triggering an alarm circuit                         |
| Expansion joints on bridges and railway lines | Small gaps allow metal to expand on hot days without buckling                                    |
| Riveting of metal plates                      | Hot rivets are inserted then cooled; they contract and pull plates tightly together              |
| Bimetallic strip in electric iron             | Regulates temperature by breaking the circuit when hot                                           |
| Sufuria lid design                            | A wider lid rim expands to fit loosely on the pot, allowing easy removal even when hot           |

### Reflection

Overhead electrical power lines are usually installed with a slight sag between poles rather than pulled tight. Why do you think engineers design them this way, considering what you have learnt about thermal expansion?

### Key Terms

Temperature • Kelvin • Thermal expansion • Linear expansivity • Bimetallic strip • Thermocouple • Anomalous expansion of water

### Worksheet 4.1

1. Convert 37°C (normal human body temperature) to Kelvin.
2. Explain, with a diagram, how a bimetallic strip thermostat switches off an electric iron when it gets too hot.
3. Describe an experiment you could carry out to show that different metals have different linear expansivities.
4. Explain why concrete pavements are constructed with small gaps between slabs.
5. Research and present: Why do plastic containers with hot liquid sometimes deform, while glass containers may crack under sudden temperature changes?

### End of Topic 4 Questions

1. Define temperature.
2. State the SI unit of temperature and convert 25°C to Kelvin.
3. Name three devices used to measure temperature and state the principle each uses.
4. Describe the unusual expansion of water between 0°C and 4°C.
5. Explain two applications of thermal expansion in everyday life.
6. Why are railway lines built with small gaps between sections?

## Topic 5: Moments and Equilibrium

Strand 1.5 • Suggested lessons: 15

### By the end of this topic, you should be able to:

- Determine the centre of gravity of regular and irregular objects
- Identify the states of equilibrium in bodies
- Determine the moment of a force about a point
- Verify the principle of moments in the turning of objects
- Describe torque and couple in turning objects
- Appreciate the applications of moments and stability in day-to-day life

*Key Inquiry Question: How does the stability of bodies affect the design of their structures?*

### 5.1 Centre of Gravity

The **centre of gravity (C.O.G.)** of a body is the point at which the entire weight of the body appears to act. For a uniform, symmetrical object such as a ruler, the centre of gravity is at its geometric centre. For irregularly shaped objects, the centre of gravity can be found experimentally using a plumb line: the object is suspended freely from different points, and a plumb line is dropped from each suspension point; the intersection of the lines gives the centre of gravity.

### 5.2 States of Equilibrium

| State                | Description                                                                                                                    | Example                                      |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Stable equilibrium   | A small displacement returns the object to its original position; centre of gravity rises when disturbed                       | A cone resting on its base                   |
| Unstable equilibrium | A small displacement causes the object to move further away from its original position; centre of gravity falls when disturbed | A cone balanced on its tip                   |
| Neutral equilibrium  | The object stays in its new position after being displaced; height of centre of gravity remains the same                       | A ball or cylinder rolling on a flat surface |

#### Idea Box

An object is more stable when it has a **low centre of gravity** and a **wide base**. This is why racing cars are built low to the ground, and why a tall, narrow lorry is more likely to topple than a low, wide one.

### 5.3 Moment of a Force

A **moment** (or turning effect) of a force is the product of the force and the perpendicular distance from the pivot (fulcrum) to the line of action of the force.

$$\text{Moment} = \text{Force} \times \text{perpendicular distance from pivot}$$

The SI unit of moment is the newton metre (Nm).

### 5.4 The Principle of Moments

The **Principle of Moments** states that for a body in equilibrium (balanced), the sum of clockwise moments about any point equals the sum of anticlockwise moments about the same point.

$$\text{Sum of clockwise moments} = \text{Sum of anticlockwise moments}$$

#### Worked Example

A uniform beam is balanced on a pivot. A 40 N weight is placed 0.3 m to the left of the pivot. How far from the pivot on the right side should a 60 N weight be placed to balance the beam?

Clockwise moment = Anticlockwise moment

$$60 \times d = 40 \times 0.3$$

$$d = 12/60 = \mathbf{0.2\ m}$$

## 5.5 Torque and Couple

A **couple** is a pair of equal and opposite forces acting on a body but not along the same line, producing pure rotation without translation (e.g. turning a steering wheel or a tap with two hands). The turning effect (moment) produced by a couple is called **torque**.

Torque = one Force  $\times$  perpendicular distance between the forces

## 5.6 Resolution of Forces

A single force can be split (resolved) into two perpendicular components – a horizontal component and a vertical component. This is useful when analysing forces acting at an angle, such as a rope pulling a sledge at an angle to the ground.

### Case Study: Stability of Matatus and Trucks

Public transport regulators in Kenya have set limits on how high goods can be loaded onto lorries and matatus. Loading goods too high raises the vehicle's centre of gravity, making it less stable and more likely to overturn while cornering, especially on Kenya's winding highways. Understanding the principle of stability (low centre of gravity + wide base) has directly informed road safety regulations such as the National Transport and Safety Authority (NTSA) rules on load height and distribution.

### Reflection

Why do tightrope walkers often carry a long pole while walking on the rope? Relate your answer to centre of gravity and stability.

### Key Terms

Centre of gravity • Equilibrium (stable/unstable/neutral) • Moment • Principle of Moments • Torque • Couple • Resolution of forces

### Worksheet 5.1

1. A see-saw of length 4 m is pivoted at its centre. A boy of weight 400 N sits 1.5 m from the pivot. Where should a girl of weight 300 N sit to balance the see-saw?
2. Using a plumb line and a piece of irregularly-shaped cardboard, describe step-by-step how you would determine its centre of gravity.
3. Give three examples each of objects in stable, unstable and neutral equilibrium found around your school compound.
4. Explain, using a diagram, how a spanner applies the principle of moments when loosening a tight nut.

### End of Topic 5 Questions

1. Define centre of gravity.
2. Distinguish between stable, unstable and neutral equilibrium.
3. State the Principle of Moments.
4. Define torque and state its SI unit.
5. A uniform metre rule is balanced at the 50 cm mark. A mass producing a force of 2 N hangs at the 20 cm mark. At what mark should a 4 N force be hung to balance the rule?
6. Explain two factors that determine the stability of an object.
7. Distinguish between a moment and a couple.

## Topic 6: Energy, Work, Power and Machines

Strand 1.6 • Suggested lessons: 18

**By the end of this topic, you should be able to:**

- Explain the meaning of energy, work and power in relation to machines
- Demonstrate the transformation of mechanical energy using simple apparatus
- Describe applications of simple machines in making work easier
- Appreciate the applications of machines in day-to-day life

*Key Inquiry Question: How do machines make work easier?*

### 6.1 Work, Energy and Power

**Work** is done when a force moves an object through a distance in the direction of the force.

$$\text{Work Done (W)} = \text{Force (F)} \times \text{Distance (d)}$$

The SI unit of work is the joule (J), where  $1 \text{ J} = 1 \text{ N} \times 1 \text{ m}$ .

**Energy** is the capacity to do work. It exists in many forms including kinetic, potential, chemical, heat, electrical, sound, light and nuclear energy.

**Power** is the rate of doing work, i.e. work done per unit time.

$$\text{Power (P)} = \text{Work Done (W)} \div \text{Time (t)}$$

The SI unit of power is the watt (W), where  $1 \text{ W} = 1 \text{ J/s}$ .

### 6.2 Kinetic and Potential Energy

$$\text{Kinetic Energy} = \frac{1}{2}mv^2$$

**Kinetic energy (K.E.)** is the energy possessed by a moving object due to its mass (m) and velocity (v).

$$\text{Gravitational Potential Energy} = mgh$$

**Gravitational potential energy (P.E.)** is the energy possessed by an object due to its position (height, h) above a reference level.

**Elastic potential energy** is the energy stored in a stretched or compressed elastic object, such as a spring or a stretched bow.

### 6.3 Law of Conservation of Energy

The **Law of Conservation of Energy** states that energy cannot be created or destroyed, only transformed from one form to another; the total energy of an isolated system remains constant.

#### Worked Example

A ball of mass 0.5 kg is thrown upward and reaches a maximum height of 4 m. Assuming no air resistance, find the speed with which the ball was thrown. ( $g = 10 \text{ m/s}^2$ )

At maximum height, all kinetic energy has converted to potential energy:

$$\frac{1}{2}mv^2 = mgh \rightarrow v^2 = 2gh = 2 \times 10 \times 4 = 80$$

$$v = \sqrt{80} = \mathbf{8.9 \text{ m/s (approx.)}}$$

### Idea Box

A swinging pendulum continuously transforms energy between kinetic energy (fastest at the bottom of the swing) and potential energy (greatest at the highest points of the swing). A bow and arrow stores elastic potential energy when drawn, which converts to kinetic energy of the arrow on release.

## 6.4 Machines

A **machine** is any device that makes work easier by changing the size, direction, or speed of an applied force. Machines do not reduce the amount of work done; they simply make it more convenient.

| Term                        | Definition / Formula                                                        |
|-----------------------------|-----------------------------------------------------------------------------|
| Effort (E)                  | The force applied to the machine                                            |
| Load (L)                    | The force to be overcome or the weight to be moved                          |
| Mechanical Advantage (M.A.) | $M.A. = \text{Load} \div \text{Effort}$                                     |
| Velocity Ratio (V.R.)       | $V.R. = \text{Distance moved by effort} \div \text{Distance moved by load}$ |
| Efficiency                  | $\text{Efficiency} = (M.A. \div V.R.) \times 100\%$                         |

### Idea Box

No real machine is 100% efficient because some input energy is always lost, usually as heat due to friction between moving parts. Efficiency is always less than 100% in practice.

## 6.5 Types of Simple Machines

- **Levers** - rigid bars that pivot about a fulcrum (e.g. wheelbarrow, pliers, bottle opener, crowbar)
- **Inclined plane** - a sloped surface used to raise loads with less effort (e.g. a ramp)
- **Pulleys** - wheels with grooves that change the direction and/or magnitude of a force (e.g. used to lift water from wells, flag hoists)
- **Wheel and axle** - a wheel attached to a smaller axle (e.g. a steering wheel, doorknob, screwdriver)
- **Gears** - toothed wheels that transmit rotational force (e.g. bicycle gears, clocks)
- **Hydraulic lift** - uses fluid pressure to multiply force (e.g. car jacks, garage lifts)
- **Pulley belt** - transmits rotational motion between two pulleys (e.g. in motor vehicle engines, sewing machines)
- **The screw** - an inclined plane wrapped around a cylinder (e.g. bolts, screw jacks)

## 6.6 Applications of Machines in Modern Technology

Simple machines are combined to build complex machines such as **treadmills** (belt and pulley systems), **elevators** (pulleys, motors, counterweights), **escalators** (chain and gear systems) and **excavators** (hydraulic systems combined with levers).

### Case Study: The Traditional Wheelbarrow vs the Modern Excavator

A simple wheelbarrow used by a Kenyan construction worker uses the principle of a lever (a class 2 lever, with the load between the effort and the fulcrum) to move sand and stones with less effort than carrying them by hand. A modern excavator used in road construction (such as along the SGR route) combines hydraulic lifts (Pascal's Principle), levers (the digging arm) and gears in its engine, showing how basic Physics principles you have learnt scale up into powerful modern machinery.

### Reflection

Some people say "machines do the work for us." Is this statement scientifically accurate? Explain your answer using the concepts of work, effort and mechanical advantage.

### Key Terms

Work • Energy • Power • Kinetic energy • Potential energy • Conservation of energy • Machine • Mechanical advantage • Velocity ratio • Efficiency

### Worksheet 6.1

1. A porter pushes a loaded cart with a force of 150 N through a distance of 20 m. Calculate the work done.
2. A machine requires an effort of 50 N to lift a load of 250 N. Calculate its mechanical advantage.

3. If the effort in question 2 moves 2 m while the load moves 0.5 m, calculate the velocity ratio and efficiency of the machine.
4. Using locally available materials (sticks, string, bottle tops), construct a simple lever or pulley system and demonstrate how it makes lifting a small load easier.
5. A stone of mass 2 kg is raised to a height of 5 m. Calculate its gravitational potential energy. ( $g = 10 \text{ m/s}^2$ )

#### End of Topic 6 Questions

1. Define work, energy and power, stating the SI unit for each.
2. State the Law of Conservation of Energy.
3. Write the formula for mechanical advantage, velocity ratio and efficiency of a machine.
4. Explain why the efficiency of a real machine is always less than 100%.
5. Name four types of simple machines and give one example of each from daily life.
6. A crane lifts a load of 500 N through a height of 6 m in 10 seconds. Calculate the power developed by the crane.
7. Describe the energy transformations that take place as a pendulum swings from one side to the other.

# STRAND 2: WAVES AND OPTICS

Topics 7-8: Properties of Waves, Radioactivity and Stability of Isotopes

# Topic 7: Properties of Waves

Strand 2.1 • Suggested lessons: 24

## By the end of this topic, you should be able to:

- Explain wave properties in real-life situations
- Demonstrate the properties of waves
- Demonstrate the formation and properties of stationary waves
- Describe applications of stationary waves
- Describe the Doppler effect and its applications
- Appreciate the formation and application of waves in real-life situations

*Key Inquiry Questions: (1) How do you relate waves to the basic properties of light? (2) Where is Doppler's effect applied in real-life situations?*

## 7.1 What is a Wave?

A **wave** is a disturbance that transfers energy from one point to another without transferring matter. Waves are classified as:

- **Transverse waves** – particles vibrate perpendicular to the direction of wave travel (e.g. water waves, light waves)
- **Longitudinal waves** – particles vibrate parallel to the direction of wave travel (e.g. sound waves)

Key wave terms: **wavelength** ( $\lambda$ ) is the distance between two successive identical points on a wave; **frequency** ( $f$ ) is the number of complete waves passing a point per second (measured in hertz, Hz); **amplitude** is the maximum displacement from the rest position; **wave speed** ( $v$ ) is related to frequency and wavelength by:

$$v = f\lambda$$

## 7.2 Properties of Waves (Qualitative)

| Property                | Description                                                                                       |
|-------------------------|---------------------------------------------------------------------------------------------------|
| Rectilinear propagation | Waves (especially light) travel in straight lines in a uniform medium                             |
| Reflection              | A wave bounces off a surface it cannot pass through, obeying the law of reflection                |
| Refraction              | A wave changes speed and direction when passing from one medium into another of different density |
| Diffraction             | A wave spreads out after passing through a narrow gap or around an obstacle                       |
| Interference            | Two waves meeting at a point combine to produce a resultant wave (constructive or destructive)    |

### Idea Box

You can observe diffraction of water waves in a harbour: waves entering through a narrow harbour mouth spread out in circular patterns once inside. You can observe interference when two stones are dropped into a still pond close together – the ripples cross and produce patterns of larger and smaller waves.

## 7.3 Applications of Wave Properties: Modulation

Radio and television signals are transmitted using electromagnetic waves. Since sound and information signals have low frequencies that do not travel far, they are combined with a high-frequency **carrier wave** through a process called **modulation**. In **Frequency Modulation (FM)**, the frequency of the carrier wave is varied according to the information signal, while its amplitude stays constant. FM broadcasting gives clearer sound with less interference from noise compared to Amplitude Modulation (AM).

## 7.4 Stationary (Standing) Waves

A **stationary wave** is formed when two waves of the same frequency and amplitude travelling in opposite directions superpose (overlap). Unlike a travelling wave, a stationary wave does not transfer energy from one point to another; instead, it has fixed points of no vibration called **nodes** and points of maximum vibration called **antinodes**.

Stationary waves are formed in vibrating strings (as in a guitar) and vibrating air columns (as in a flute or organ pipe). When the frequency of an external force matches the natural frequency of an object, **resonance** occurs, producing a stationary wave of large amplitude. This is used in frequency-modulated wave detection circuits (resonant circuits select a specific broadcast frequency).

#### Idea Box

Resonance explains why a radio can be "tuned" to a particular station: the receiver circuit is adjusted until its natural frequency matches (resonates with) the frequency of the desired broadcast signal.

## 7.5 The Doppler Effect

The **Doppler effect** is the apparent change in frequency (and pitch, for sound) of a wave due to relative motion between the source and the observer. When a source moves towards an observer, the observed frequency increases (sound appears higher pitched); when the source moves away, the observed frequency decreases (sound appears lower pitched).

#### Case Study: The Ambulance Siren

As an ambulance approaches you along a busy Nairobi street, its siren sounds higher pitched. As it passes and moves away, the pitch suddenly drops. This is the Doppler effect in action: the sound waves are compressed in front of the moving ambulance (higher frequency) and stretched out behind it (lower frequency). The Doppler effect is also used by traffic police radar guns to measure vehicle speed, and by astronomers to determine whether distant stars and galaxies are moving toward or away from Earth (red-shift and blue-shift).

#### Reflection

Why do you think FM radio reception is generally clearer in hilly or built-up areas compared to AM radio? Think about how each type of wave interacts with obstacles.

#### Key Terms

Wave • Transverse wave • Longitudinal wave • Wavelength • Frequency • Amplitude • Diffraction • Interference • Modulation • Stationary wave • Resonance • Doppler effect

#### Worksheet 7.1

1. A wave has a frequency of 50 Hz and a wavelength of 4 m. Calculate its speed.
2. Sketch and label a transverse wave, showing the wavelength and amplitude.
3. Using a rope tied to a wall, demonstrate to your group how a stationary wave forms when you shake the free end at the right frequency.
4. Explain, using the Doppler effect, why the sound of a passing train changes pitch as it moves past a platform.

#### End of Topic 7 Questions

1. Distinguish between a transverse wave and a longitudinal wave, giving one example of each.
2. Define wavelength, frequency and amplitude.
3. State the wave equation relating speed, frequency and wavelength.
4. Explain the difference between a travelling wave and a stationary wave.
5. Define resonance and give one everyday example.
6. State the Doppler effect and describe one practical application.
7. A radio wave travels at  $3 \times 10^8$  m/s and has a frequency of 100 MHz. Calculate its wavelength.

# Topic 8: Radioactivity and Stability of Isotopes

Strand 2.2 • Suggested lessons: 24

## By the end of this topic, you should be able to:

- Explain the terminologies used in radioactivity
- Identify the types and properties of radioactive emissions
- Illustrate how radionuclides attain stability using nuclear equations
- Demonstrate the detection of radioactive emissions
- Determine the half-life of common radioactive elements
- Appreciate the applications and safety precautions of radioactivity

*Key Inquiry Questions: (1) How is radioactivity important in day-to-day life? (2) What are the risks of exposure to radiation?*

## 8.1 Key Terms in Radioactivity

- **Nuclide** – a species of atom with a specific number of protons and neutrons
- **Isotopes** – atoms of the same element with the same number of protons but different numbers of neutrons
- **Radioisotope** – an unstable isotope that emits radiation as it decays
- **Radioactivity** – the spontaneous emission of radiation from unstable atomic nuclei
- **Background radiation** – low-level radiation that is always present in the environment from natural sources such as rocks, cosmic rays and soil

## 8.2 Types and Properties of Radioactive Emissions

| Radiation          | Nature                                 | Charge      | Penetrating Power                         | Ionizing Power |
|--------------------|----------------------------------------|-------------|-------------------------------------------|----------------|
| Alpha ( $\alpha$ ) | Helium nucleus (2 protons, 2 neutrons) | +2          | Low (stopped by paper/skin)               | Very high      |
| Beta ( $\beta$ )   | Fast-moving electron                   | -1          | Medium (stopped by a few mm of aluminium) | Medium         |
| Gamma ( $\gamma$ ) | High-energy electromagnetic wave       | 0 (neutral) | Very high (needs thick lead/concrete)     | Low            |

### Idea Box

Remember: the more penetrating a radiation is, the less ionizing it tends to be, and vice versa. Alpha particles are the most dangerous if swallowed or inhaled (due to high ionizing power at close range) but the easiest to shield against from outside the body.

## 8.3 Nuclear Equations and Stability

When a nucleus emits radiation, it changes into a different nuclide. This can be represented using nuclear equations that balance mass number (top) and atomic number (bottom):

- Alpha decay: mass number decreases by 4, atomic number decreases by 2
- Beta decay: mass number stays the same, atomic number increases by 1 (a neutron changes into a proton and an electron)
- Gamma emission: no change in mass number or atomic number; the nucleus simply loses excess energy

Radionuclides continue emitting radiation (decaying) until they form a stable nuclide – one that does not spontaneously emit radiation.

## 8.4 Detection of Radioactive Emissions

- **Photographic film/plate** – radiation darkens photographic film, used in radiation badges worn by hospital and nuclear plant workers
- **Cloud chamber** – radiation leaves visible vapour trails as it ionizes gas particles
- **Geiger-Muller (GM) tube/counter** – radiation ionizes gas inside the tube, producing an electric pulse that is counted and often converted to an audible click
- **Gold-leaf electroscope** – radiation ionizes air around the electroscope, causing it to discharge (the leaf falls) faster than normal

## 8.5 Half-Life

The **half-life** ( $t_{1/2}$ ) of a radioactive substance is the time taken for half of the original number of radioactive atoms (or the activity) in a sample to decay.

$$N = N_0 \left(\frac{1}{2}\right)^{t/t_{1/2}}$$

where  $N$  is the number of undecayed atoms remaining after time  $t$ , and  $N_0$  is the original number of atoms.

### Worked Example

A radioactive sample has a half-life of 6 hours. If the initial activity is 800 counts per minute, what will the activity be after 18 hours?

Number of half-lives =  $18/6 = 3$

Remaining activity =  $800 \times \left(\frac{1}{2}\right)^3 = 800 \times 1/8 = \mathbf{100 \text{ counts per minute}}$

## 8.6 Nuclear Fission and Nuclear Fusion

**Nuclear fission** is the splitting of a large, heavy nucleus (e.g. Uranium-235) into two smaller nuclei, releasing a large amount of energy. This process, once started, can sustain a **chain reaction**, and is used in nuclear power stations and nuclear weapons.

**Nuclear fusion** is the joining of two light nuclei (e.g. hydrogen isotopes) to form a heavier nucleus, releasing even more energy per reaction than fission. Fusion is the process that powers the sun and other stars.

**Note:** Nuclear reactions are different from chemical reactions – nuclear reactions involve changes in the nucleus and release far more energy than chemical reactions, which only involve the electrons.

## 8.7 Applications and Safety Precautions

| Field       | Application                                                                       |
|-------------|-----------------------------------------------------------------------------------|
| Medicine    | Radiotherapy (destroying cancer cells), diagnostic imaging, sterilizing equipment |
| Industry    | Checking for cracks/flaws in metal (non-destructive testing), thickness gauges    |
| Agriculture | Pest control (sterile insect technique), soil moisture measurement                |
| Archaeology | Carbon dating – determining the age of fossils and ancient artefacts              |
| Energy      | Nuclear power generation                                                          |

**Safety precautions** when handling radioactive materials include: using lead shielding, keeping maximum distance from sources, minimizing exposure time, wearing protective clothing and radiation badges, and proper disposal of radioactive waste in sealed, labelled containers.

### Case Study: The Chernobyl Disaster

In 1986, a nuclear reactor at Chernobyl (in present-day Ukraine) exploded due to a failed safety test, releasing large amounts of radioactive material into the environment. The disaster caused immediate deaths, long-term health effects such as cancer among exposed populations, and forced the permanent evacuation of nearby towns. This tragedy highlights why strict safety precautions, proper reactor design, and international regulation of nuclear technology are essential when harnessing the benefits of nuclear fission.

### Reflection

Radioactivity has both life-saving uses (cancer treatment) and life-threatening dangers (radiation sickness, nuclear disasters). How should countries balance the benefits of nuclear technology against its risks?

### Key Terms

Radioactivity • Isotope • Radioisotope • Alpha/Beta/Gamma radiation • Half-life • Nuclear fission • Nuclear fusion • Chain reaction • Background radiation

1. Make a chart comparing the nature, charge, penetrating power and ionizing power of alpha, beta and gamma radiation.
2. A sample of a radioactive isotope has a half-life of 10 days. If the original mass is 80 g, what mass will remain after 40 days?
3. Explain why gamma radiation, despite being the least ionizing, is considered very dangerous when a person is exposed to a large source from outside the body.
4. Research and present a one-page report on how carbon dating is used to determine the age of ancient artefacts.

### End of Topic 8 Questions

1. Distinguish between an isotope and a radioisotope.
2. Name the three main types of radioactive emissions and state one property of each.
3. Define half-life.
4. Distinguish between nuclear fission and nuclear fusion.
5. State three applications of radioactivity in medicine and industry.
6. List four safety precautions taken when handling radioactive substances.
7. A radioactive source has a half-life of 4 years. Calculate the fraction of the original sample remaining after 12 years.

# STRAND 3: ELECTRICITY AND MAGNETISM

Topics 9-11: Electrostatics, Current Electricity, Introduction to Electronics

# Topic 9: Electrostatics

Strand 3.1 • Suggested lessons: 10

**By the end of this topic, you should be able to:**

- Explain the origin of charges in a material
- Describe methods of charging a conductor
- Illustrate charge distribution on conductors of various shapes
- Explain functions of the various parts of an electroscope
- Illustrate charging of an electroscope
- Appreciate uses of an electroscope and applications of static charges in day-to-day life

Key Inquiry Questions: (1) How do lightning arrestors work? (2) How do materials get charged?

## 9.1 Origin of Charge

All matter is made up of atoms, which consist of a central **nucleus** (containing positively charged **protons** and neutral **neutrons**) surrounded by negatively charged **electrons**. Normally, an atom has equal numbers of protons and electrons, making it electrically neutral. An object becomes **charged** when it gains or loses electrons: gaining electrons makes it negatively charged; losing electrons makes it positively charged. The SI unit of charge is the **coulomb (C)**.

 **Idea Box**

Law of electrostatics: **like charges repel, unlike charges attract**. This simple rule explains most electrostatic phenomena you will observe.

## 9.2 Methods of Charging a Conductor

| Method             | Description                                                                                                            |
|--------------------|------------------------------------------------------------------------------------------------------------------------|
| Friction (rubbing) | Rubbing two different materials transfers electrons from one to the other (e.g. rubbing a plastic ruler with a cloth)  |
| Contact            | Touching a charged object to a neutral conductor transfers some charge directly to it                                  |
| Induction          | Bringing a charged object near (but not touching) a conductor causes a redistribution of charge without direct contact |

## 9.3 Charge Distribution on Conductors

Charge on a conductor spreads over its outer surface. The distribution depends on the shape of the conductor: on a **sphere**, charge distributes evenly; on conductors with sharp points or edges (e.g. pear-shaped or wedge-shaped conductors), charge concentrates most densely at the sharpest points. This is why lightning arrestors are made with sharp spikes – charge concentration at the point allows it to safely discharge (leak) into the air or ground gradually, reducing the risk of a sudden, damaging lightning strike.

## 9.4 The Gold-Leaf Electroscope

An **electroscope** is an instrument used to detect the presence, estimate the amount, and identify the type (positive or negative) of electric charge on an object. It consists of a metal cap, a metal rod, and a thin gold (or aluminium) leaf attached to the rod, enclosed in a case with glass sides to prevent air currents from disturbing the leaf.

When a charged object touches the cap (charging by contact) or is brought near it (charging by induction), charge spreads to the leaf and rod. Since the leaf and rod now have the same type of charge, they repel each other, causing the leaf to rise/diverge. The extent of divergence gives a rough estimate of the amount of charge present.

 **Idea Box**

An electroscope can also be used to test whether a material is a conductor or an insulator: if touching a material to a charged electroscope causes the leaf to fall (discharge), the material is a conductor.

## 9.5 Applications of Electrostatics

- **Spray guns** – paint droplets are charged so they spread evenly and are attracted to the (oppositely charged or earthed) surface being painted, reducing wastage
- **Photocopiers and laser printers** – use charged drums to attract toner particles onto paper
- **Electrostatic precipitators** – remove dust and pollutant particles from industrial chimney smoke by charging the particles and attracting them to charged plates
- **Fingerprint detection** – static charge techniques help make latent fingerprints visible
- **Lightning and lightning arrestors** – tall buildings use pointed metal rods connected to the ground to safely discharge accumulated atmospheric charge and prevent damage from lightning strikes

### Case Study: Lightning Safety in Kenya

Parts of Kenya, especially the western and Rift Valley regions, experience frequent and severe thunderstorms, sometimes resulting in loss of life and property from lightning strikes. Understanding electrostatics has led to safety measures such as installing lightning arrestors (also called lightning rods) on tall buildings, schools and churches. These pointed metal conductors are connected to the ground by a thick cable; because charge concentrates at sharp points, they allow gradual, safe discharge of atmospheric electricity, protecting structures and lives.

### Reflection

Have you ever experienced a small electric "shock" after walking on a carpet and touching a metal door handle, or after removing a synthetic sweater? Explain what causes this using what you have learnt about static charges.

### Key Terms

Electron • Proton • Coulomb • Friction • Contact • Induction • Electroscope • Lightning arrestor • Electrostatic precipitator

### Worksheet 9.1

1. Using a plastic pen and small pieces of paper, demonstrate charging by friction. Record your observations.
2. With the aid of labelled diagrams, describe how a gold-leaf electroscope is charged by (a) contact and (b) induction.
3. Explain why lightning arrestors are made with sharp points rather than flat or rounded surfaces.
4. Construct a simple electroscope using locally available materials (a jar, wire, and foil) and test it with a charged plastic ruler.

### End of Topic 9 Questions

1. State the law of electrostatics.
2. Explain how an object becomes positively charged and how it becomes negatively charged.
3. Distinguish between charging by contact and charging by induction.
4. Describe the structure and function of a gold-leaf electroscope.
5. Explain how a lightning arrestor protects a building.
6. State two applications of electrostatic precipitators.

## Topic 10: Current Electricity

Strand 3.2 • Suggested lessons: 18

### By the end of this topic, you should be able to:

- Explain the terminologies used in current electricity
- Verify the relationships  $V=IR$  and  $E=I(R+r)$
- Determine the resistance and resistivity of various conductors
- Determine the effective resistance of resistor networks
- Determine the relationship of potential difference and current to power
- Appreciate the applications of current electricity in day-to-day life

*Key Inquiry Questions: (1) How is current electricity applicable in our day-to-day life? (2) Why are resistors used in electrical circuits?*

### 10.1 Key Terms

- **Electric current (I)** – the rate of flow of electric charge, measured in amperes (A)
- **Potential difference (V)** – the work done to move a unit charge between two points in a circuit, measured in volts (V)
- **Electromotive force (e.m.f., E)** – the total energy supplied per unit charge by a source (e.g. a cell), including energy lost inside the source itself
- **Internal resistance (r)** – the resistance offered by the materials inside a cell/battery to the flow of current
- **Resistance (R)** – the opposition to the flow of current, measured in ohms ( $\Omega$ )

### 10.2 Ohm's Law

**Ohm's Law** states that the current flowing through a metallic conductor is directly proportional to the potential difference across it, provided temperature and other physical conditions remain constant.

$$V = IR$$

A conductor that obeys Ohm's Law (produces a straight-line graph of V against I through the origin) is called an **ohmic conductor** (e.g. a metal wire at constant temperature). A conductor that does not obey Ohm's Law is called **non-ohmic** (e.g. a filament lamp, whose resistance increases as it heats up, or a diode).

### 10.3 E.M.F., Terminal Voltage and Internal Resistance

When a cell drives current through a circuit, some energy is used up overcoming the cell's own internal resistance. This gives the relationship:

$$E = I(R + r)$$

where E is the e.m.f. of the cell, I is the current, R is the external resistance, and r is the internal resistance of the cell.

#### Worked Example

A cell of e.m.f. 6 V and internal resistance  $0.5 \Omega$  is connected to an external resistor of  $5.5 \Omega$ . Calculate the current flowing in the circuit.

$$E = I(R + r) \rightarrow 6 = I(5.5 + 0.5) = I(6)$$

$$I = 6/6 = \mathbf{1 \text{ A}}$$

### 10.4 Resistance and Resistivity

The resistance of a conductor depends on: its length (directly proportional), its cross-sectional area (inversely proportional), the material it is made of, and its temperature. Resistance can be determined experimentally using:

- **Resistor colour codes** – bands of colour on a resistor indicate its resistance value

- **Ammeter-voltmeter method** – measuring current and voltage and applying  $R = V/I$
- **Wheatstone bridge / Metre bridge** – a balanced bridge circuit used to accurately compare an unknown resistance with known resistances

## 10.5 Resistor Networks

| Arrangement | Effective Resistance Formula                         |
|-------------|------------------------------------------------------|
| Series      | $R_{\text{total}} = R_1 + R_2 + R_3 + \dots$         |
| Parallel    | $1/R_{\text{total}} = 1/R_1 + 1/R_2 + 1/R_3 + \dots$ |

### Worked Example

Two resistors of 4 Ω and 6 Ω are connected: (a) in series, (b) in parallel. Find the effective resistance in each case.

- (a) Series:  $R = 4 + 6 = \mathbf{10\ \Omega}$   
 (b) Parallel:  $1/R = 1/4 + 1/6 = 3/12 + 2/12 = 5/12 \rightarrow R = 12/5 = \mathbf{2.4\ \Omega}$

### Idea Box

In series connections, the total resistance is always greater than any individual resistor. In parallel connections, the total resistance is always smaller than the smallest individual resistor.

## 10.6 Electrical Power and Heating Effect

Power (P) = VI = I²R = V²/R

Electrical energy is converted into heat as current flows through a resistor – this is the **heating effect of an electric current**, applied in electric heaters, kettles, cookers, irons and incandescent light bulbs.

### Case Study: Household Electrical Wiring and Safety

In Kenyan homes connected to the national grid (supplied by Kenya Power), appliances are wired in parallel so that each device (light bulb, TV, fridge) receives the full mains voltage and can be switched on or off independently without affecting others. Fuses and circuit breakers are placed in series in the main circuit to protect against excessive current (which could cause overheating and fire) by breaking the circuit when current exceeds a safe limit. Understanding resistance, current and power calculations ( $P = VI$ ) allows electricians to correctly size fuses and wiring for the load in a building.

### Reflection

Why do you think Christmas tree lights are wired in parallel rather than in series today, when older designs used series wiring (where one blown bulb would turn off the whole string)?

### Key Terms

Current • Potential difference • E.M.F. • Internal resistance • Ohm's Law • Ohmic/Non-ohmic conductor • Resistivity • Series • Parallel • Electrical power

### Worksheet 10.1

1. A conductor carries a current of 2 A when a potential difference of 12 V is applied across it. Calculate its resistance.
2. Three resistors of 2 Ω, 3 Ω and 5 Ω are connected in series to a 20 V battery. Calculate (a) the total resistance, (b) the current flowing.
3. Draw a circuit diagram showing two resistors connected in parallel with an ammeter and a voltmeter correctly positioned.
4. An electric kettle is rated 240 V, 2000 W. Calculate the current it draws when operating normally.
5. In groups, use a cell, ammeter, voltmeter, rheostat and a resistance wire to plot a graph of V against I and verify Ohm's Law.

### End of Topic 10 Questions

1. State Ohm's Law.
2. Distinguish between e.m.f. and terminal potential difference.
3. Write the formula relating e.m.f., current, external resistance and internal resistance.
4. State two factors that affect the resistance of a conductor.
5. Two resistors of  $6\ \Omega$  and  $12\ \Omega$  are connected in parallel. Calculate the effective resistance.
6. Define electrical power and state its SI unit.
7. Explain why a fuse is always connected in series in a circuit, never in parallel.

# Topic 11: Introduction to Electronics

Strand 3.3 • Suggested lessons: 6

## By the end of this topic, you should be able to:

- Explain the meaning of insulator, conductor, semiconductor and superconductor
- Distinguish between insulators, conductors, semiconductors and superconductors
- Investigate the electrical behaviour of these materials with varying temperature
- Explain intrinsic and extrinsic semiconductors
- Explain the formation of n-type and p-type semiconductors
- Describe applications of these materials in day-to-day life

*Key Inquiry Questions: (1) How does temperature affect the resistance of conductors and semiconductors? (2) What is the significance of semiconductors in day-to-day life?*

## 11.1 Conductors, Insulators, Semiconductors and Superconductors

| Material       | Description                                                                                                       | Examples                                                            |
|----------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| Conductor      | Allows electric current to flow easily due to many free electrons                                                 | Copper, aluminium, silver                                           |
| Insulator      | Does not allow current to flow easily; very few free electrons                                                    | Rubber, glass, plastic, dry wood                                    |
| Semiconductor  | Has electrical conductivity between that of a conductor and an insulator; conductivity increases with temperature | Silicon, germanium                                                  |
| Superconductor | Offers zero electrical resistance below a very low critical temperature                                           | Certain metals and alloys at extremely low (cryogenic) temperatures |

### Idea Box

Metals (conductors) become MORE resistive as temperature increases (more particle vibration disrupts electron flow). Semiconductors behave oppositely – their resistance DECREASES as temperature increases, because more electrons gain enough energy to become free charge carriers.

## 11.2 Energy Band Theory

In energy band theory, electrons in a solid occupy energy levels grouped into bands: the **valence band** (occupied by outer electrons) and the **conduction band** (where electrons are free to move and conduct current), separated by an **energy gap**.

- In **conductors**, the valence and conduction bands overlap, so electrons move freely
- In **insulators**, there is a very large energy gap, so electrons cannot easily jump to the conduction band
- In **semiconductors**, the energy gap is small, so some electrons can cross it, especially when heated or given extra energy

## 11.3 Intrinsic and Extrinsic Semiconductors

A **pure semiconductor** (e.g. pure silicon) is called an **intrinsic semiconductor**. It has few free charge carriers at room temperature, giving it low conductivity.

To improve conductivity for practical use, a small amount of impurity is deliberately added to the pure semiconductor in a process called **doping**, producing an **extrinsic semiconductor**:

| Type   | Impurity Added                                                    | Majority Charge Carriers           |
|--------|-------------------------------------------------------------------|------------------------------------|
| n-type | Pentavalent atoms (5 valence electrons, e.g. phosphorus, arsenic) | Free electrons (negative)          |
| p-type | Trivalent atoms (3 valence electrons, e.g. boron, gallium)        | "Holes" (positive charge carriers) |

n-type and p-type semiconductors are joined to form **p-n junctions**, which are the basis of diodes, transistors and integrated circuits used in virtually all modern electronic devices.

## 11.4 Applications

| Material        | Applications                                                                               |
|-----------------|--------------------------------------------------------------------------------------------|
| Conductors      | Electrical wiring, cables, motor windings                                                  |
| Insulators      | Cable coatings, plug casings, handles of electrical tools                                  |
| Semiconductors  | Diodes, transistors, solar cells, integrated circuits (microchips) in phones and computers |
| Superconductors | MRI machines, maglev trains, particle accelerators (require extreme cooling)               |

 **Case Study: Solar Power and Semiconductors in Kenya**

Off-grid communities in northern and rural Kenya increasingly rely on solar panels, which are made of semiconductor material (usually silicon), to generate electricity where the national grid has not reached. Sunlight striking the p-n junction in a solar cell generates a potential difference (the photovoltaic effect), driving current through a circuit. This technology, rooted in the same semiconductor physics you have learnt, has expanded access to lighting, phone charging and refrigeration in remote parts of the country.

 **Reflection**

Almost every modern device (phones, computers, TVs, solar inverters) depends on semiconductors. What do you think would happen to modern life if semiconductors had never been discovered or developed?

**Key Terms**

Conductor • Insulator • Semiconductor • Superconductor • Energy band • Doping • Intrinsic/Extrinsic semiconductor • n-type • p-type • p-n junction

 **Worksheet 11.1**

- Classify the following as conductor, insulator, or semiconductor: copper wire, rubber glove, silicon chip, glass rod, aluminium cable, germanium diode.
- Explain, using energy band theory, why insulators do not conduct electricity easily at room temperature.
- Distinguish between an n-type and a p-type semiconductor.
- Research and write a short paragraph on how a solar cell converts sunlight into electricity.

 **End of Topic 11 Questions**

- Define the terms conductor, insulator and semiconductor.
- How does an increase in temperature affect the resistance of (a) a metal conductor, (b) a semiconductor?
- What is doping? Why is it done to pure semiconductors?
- Distinguish between an intrinsic and an extrinsic semiconductor.
- Name two applications each of semiconductors and superconductors.
- Explain briefly what a superconductor is and state one condition required for superconductivity to occur.

# STRAND 4: ENVIRONMENTAL AND SPACE PHYSICS

Topics 12-13: Greenhouse Effect and Climate Change, Introduction to Space Physics

# Topic 12: Greenhouse Effect and Climate Change

Strand 4.1 • Suggested lessons: 5

**By the end of this topic, you should be able to:**

- Explain the greenhouse effect and climate change in the environment
- Outline the factors leading to the greenhouse effect
- Explain the effect of the ozone layer on climate change
- Describe the mitigating factors against climate change
- Appreciate the impact of climate change on the environment

*Key Inquiry Questions: (1) How do human actions impact climate change? (2) How does ozone layer depletion threaten our environment?*

## 12.1 The Greenhouse Effect

The **greenhouse effect** is the natural process by which certain gases in the Earth's atmosphere (**greenhouse gases**) trap heat from the sun, keeping the Earth warm enough to support life. Sunlight passes through the atmosphere and warms the Earth's surface; the surface then radiates heat back as infrared radiation, some of which is absorbed and re-emitted by greenhouse gases instead of escaping into space. Without any greenhouse effect, the Earth would be too cold to support life as we know it.

**Idea Box**

You can demonstrate the greenhouse effect concept using two closed cars parked in the sun: one with windows closed becomes noticeably hotter inside than the surrounding air, because sunlight enters through the glass and heat becomes trapped – similar to how greenhouse gases trap heat in the atmosphere.

## 12.2 Greenhouse Gases

| Gas                               | Main Source                                                       |
|-----------------------------------|-------------------------------------------------------------------|
| Carbon dioxide (CO <sub>2</sub> ) | Burning fossil fuels, deforestation, vehicle emissions            |
| Methane (CH <sub>4</sub> )        | Livestock digestion, rice paddies, decomposing waste in landfills |
| Nitrous oxide (N <sub>2</sub> O)  | Fertilizers, industrial processes                                 |
| Water vapour                      | Natural evaporation (most abundant greenhouse gas)                |
| Chlorofluorocarbons (CFCs)        | Old refrigerants, aerosol sprays                                  |

## 12.3 Global Warming and Climate Change

**Global warming** is the gradual increase in the Earth's average surface temperature, mainly due to increased concentration of greenhouse gases from human activities (burning fossil fuels, deforestation, industrial processes). This leads to **climate change** – long-term shifts in temperature and weather patterns, including changing rainfall patterns, more frequent droughts and floods, and rising sea levels.

## 12.4 Role of the Ozone Layer

The **ozone layer** is a region of the upper atmosphere (stratosphere) rich in ozone (O<sub>3</sub>) molecules, which absorb most of the sun's harmful ultraviolet (UV) radiation, protecting living organisms on Earth. Certain chemicals, especially chlorofluorocarbons (CFCs) previously used in refrigerators, air conditioners and aerosol sprays, were found to destroy ozone molecules, thinning the ozone layer (the "ozone hole") and allowing more harmful UV radiation to reach the Earth's surface. This increases risks of skin cancer, cataracts, and damage to crops and marine life.

**Idea Box**

The international Montreal Protocol (1987) banned the production of most ozone-depleting substances. Scientific monitoring shows the ozone layer has been gradually recovering since then – a rare example of successful global environmental cooperation.

## 12.5 Mitigating Climate Change

- Switching to renewable energy sources (solar, wind, geothermal, hydro)
- Afforestation and reforestation (planting more trees to absorb CO<sub>2</sub>)
- Reducing deforestation and forest degradation
- Improving energy efficiency in industries, vehicles and homes
- Promoting public transport and reducing reliance on fossil-fuel vehicles
- Proper waste management (reducing methane from landfills)
- International agreements and policies (e.g. the Paris Agreement)

### Case Study: Kenya's Geothermal Energy and the Green Belt Movement

Kenya generates a significant share of its electricity from geothermal power (from the Olkaria fields in the Rift Valley), a clean, renewable source that produces far fewer greenhouse gas emissions than coal or oil-fired power plants. Additionally, the Green Belt Movement, founded by the late Nobel Laureate Prof. Wangari Maathai, has mobilised communities across Kenya to plant millions of trees, helping to absorb carbon dioxide, restore degraded land, and demonstrate grassroots action against climate change.

### Reflection

What changes can you, your family, and your school make in daily life to reduce your contribution to greenhouse gas emissions?

### Key Terms

Greenhouse effect • Greenhouse gas • Global warming • Climate change • Ozone layer • Ultraviolet radiation • Montreal Protocol

### Worksheet 12.1

1. List five human activities that increase greenhouse gas emissions in your local area.
2. Explain, in your own words, how the greenhouse effect keeps the Earth warm.
3. Design a poster illustrating three ways your school can reduce its carbon footprint.
4. Research and report on one visible effect of climate change in your county (e.g. changing rainfall patterns, drought, flooding).

### End of Topic 12 Questions

1. Define the greenhouse effect.
2. Name three greenhouse gases and their main sources.
3. Distinguish between global warming and climate change.
4. Explain the importance of the ozone layer to life on Earth.
5. State four measures that can be taken to mitigate climate change.
6. Explain how deforestation contributes to climate change.

# Topic 13: Introduction to Space Physics

Strand 4.2 • Suggested lessons: 6

**By the end of this topic, you should be able to:**

- Describe the Big Bang theory of the origin of the universe
- Classify celestial bodies in the universe
- Outline the evolution of astrophysics and space exploration
- Explain the motion of planets around the sun
- Appreciate the careers in space exploration

Key Inquiry Questions: (1) How was the universe/earth formed? (2) How do we benefit from Astrophysics?

## 13.1 The Big Bang Theory

The **Big Bang Theory** is the leading scientific explanation for the origin of the universe. It states that the universe began about 13.8 billion years ago from an extremely hot, dense point, which then expanded rapidly and has continued expanding and cooling ever since, eventually forming galaxies, stars and planets. Evidence supporting the Big Bang theory includes the observed expansion of the universe (galaxies moving away from each other) and the detection of cosmic microwave background radiation, considered a "afterglow" of the initial explosion.

## 13.2 Classification of Celestial Bodies

| Body                     | Description                                                                                           |
|--------------------------|-------------------------------------------------------------------------------------------------------|
| Star                     | A massive, hot ball of gas that produces its own light and heat through nuclear fusion (e.g. the Sun) |
| Planet                   | A large body that orbits a star and does not produce its own light (e.g. Earth, Mars)                 |
| Moon (natural satellite) | A body that orbits a planet (e.g. the Earth's Moon)                                                   |
| Asteroid                 | A small, rocky body orbiting the sun, mostly found in the asteroid belt between Mars and Jupiter      |
| Comet                    | An icy body that develops a glowing tail as it approaches the sun and its ice vaporises               |
| Galaxy                   | A huge collection of billions of stars, gas and dust bound together by gravity (e.g. the Milky Way)   |

## 13.3 Planetary Motion

Planets orbit the sun in elliptical (oval-shaped) paths due to the sun's gravitational pull, as described by Kepler's Laws of planetary motion and later explained by Newton's Law of Universal Gravitation. The order of the planets from the sun is: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune.

 **Idea Box**

A planet closer to the sun (like Mercury) orbits faster and completes its year more quickly than a planet further away (like Neptune), because the sun's gravitational pull is stronger at shorter distances.

## 13.4 Telescopy and Space Exploration

A **telescope** is an instrument used to observe distant celestial objects by collecting and focusing light (optical telescopes) or other forms of electromagnetic radiation (e.g. radio telescopes). The history of astrophysics has evolved from early naked-eye observations by ancient civilizations, to Galileo's refracting telescope in the 1600s, to modern space telescopes such as the Hubble Space Telescope and the James Webb Space Telescope, which orbit above the Earth's atmosphere for a clearer view of the universe.

Space exploration milestones include the launch of Sputnik 1 (the first artificial satellite, 1957), the Apollo 11 Moon landing (1969), and ongoing missions to Mars and beyond using robotic rovers and probes.

## 13.5 Careers in Space Exploration

- Astronomer / Astrophysicist – studies celestial objects and the universe

- Aerospace Engineer – designs spacecraft, satellites and rockets
- Astronaut – travels into space to conduct research and missions
- Satellite Communications Technician – maintains satellite systems used for communication, weather forecasting and GPS
- Remote Sensing / GIS Specialist – uses satellite data for mapping, agriculture and disaster management

### **Case Study: Kenya's Journey into Space Science**

Kenya has been active in space science since the 1960s and 1970s, when the San Marco equatorial launch platform off the coast of Malindi was used by Italy for satellite launches, taking advantage of Kenya's position near the equator (ideal for efficient rocket launches). More recently, Kenya launched its first operational satellite, Taifa-1, in 2023, to collect data supporting agriculture, environmental monitoring, and resource management. This shows how space physics knowledge translates into practical national development.

### **Reflection**

Why do you think studying distant stars and galaxies (which may seem far removed from daily life) is still considered valuable enough for countries to invest billions of shillings in space programmes?

### **Key Terms**

Big Bang Theory • Star • Planet • Galaxy • Asteroid • Comet • Telescope • Satellite • Astrophysics

### **Worksheet 13.1**

1. Draw and label a simple diagram of the solar system showing the order of the planets from the sun.
2. Explain, in your own words, the main idea behind the Big Bang Theory.
3. Using digital devices or library resources, research one space mission (past or current) and present a summary of its purpose and achievements to the class.
4. Model the motion of a planet around the sun using a ball on a string, and explain what provides the centripetal force in this model.

### **End of Topic 13 Questions**

1. State the Big Bang Theory of the origin of the universe.
2. Give two pieces of evidence supporting the Big Bang Theory.
3. Classify and describe four types of celestial bodies.
4. Explain why planets closer to the sun take a shorter time to complete one orbit than those further away.
5. Name two instruments used in space exploration and state their uses.
6. State three careers related to space physics and astronomy.

# ANSWER KEY

Suggested answers to all Worksheets, Exercises and End of Topic Questions in this book

# Answers to All Topics

Use this chapter to check your work after attempting each question on your own.

## Topic 1: Introduction to Physics

**Worksheet 1.1:** Learner-led activities and presentations – assess based on correct identification of Physics branches, well-researched career charts, and clear presentation of interview findings. No single fixed answer.

### End of Topic Questions:

1. Physics is a branch of science that studies matter, energy, and how they interact, explaining natural phenomena through observation, measurement and experiment.
2. Any four of: Mechanics, Thermodynamics, Electricity & Magnetism, Waves, Geometrical Optics, Electronics, Modern Physics, Astronomy.
3. A law describes an observed pattern in nature (often mathematically); a theory explains why that pattern occurs and has been extensively tested.
4. Any three of: irrigation pump design, greenhouse technology (temperature/light control), machinery for ploughing/harvesting, use of levers and pulleys on farms.
5. Any three of: Engineering, Medicine/Radiography, Meteorology, Astronomy, Renewable energy technology, Aviation, ICT, Research.
6. Example: Understanding of forces and levers explains how bones and muscles work together to move the body; understanding of optics explains how the human eye focuses light to form images.

## Topic 2: Pressure

### Worksheet 2.1:

1.  $\text{Area} = 0.5 \times 0.4 = 0.2 \text{ m}^2$ ;  $\text{Force} = mg = 20 \times 10 = 200 \text{ N}$ ;  $P = F/A = 200/0.2 = \mathbf{1000 \text{ Pa}}$
2.  $P = \rho gh = 13600 \times 10 \times 0.76 = \mathbf{103,360 \text{ Pa}}$
3. Dams are thicker at the bottom because water pressure increases with depth, so the dam wall must withstand greater force at the base.
4. Learner diagram/explanation: pulling the piston up reduces pressure inside the barrel below atmospheric pressure, so atmospheric pressure pushes air/liquid in through the valve; pushing the piston down compresses and expels the air/liquid through the outlet valve.
5. Practical activity – learners should observe water spurting further from holes lower down a bottle, showing pressure increases with depth.

### End of Topic Questions:

1. Pressure is force acting normally per unit area; SI unit is the pascal (Pa).
2. Depth, density of the fluid, and acceleration due to gravity.
3.  $P = \rho gh$ , where  $P$  = pressure,  $\rho$  = density,  $g$  = acceleration due to gravity,  $h$  = depth.
4. Pascal's Principle: pressure applied to an enclosed fluid is transmitted equally throughout the fluid and to the walls of its container.
5.  $P = \rho gh = 1030 \times 10 \times 10 = \mathbf{103,000 \text{ Pa}}$
6. A small force on a small-area piston creates pressure that, by Pascal's Principle, is transmitted equally to a larger piston; since  $\text{Force} = \text{Pressure} \times \text{Area}$ , the larger piston (larger area) experiences a much larger force, lifting the car.
7. Any two of: drinking through a straw, use of a syringe, functioning of a barometer/altimeter, suction of a rubber sucker.

## Topic 3: Mechanical Properties of Materials

### Worksheet 3.1:

1.  $k = F/\Delta L = 6/(0.03) = \mathbf{200 \text{ N/m}}$
2.  $\text{Stress} = F/A = 100/(2.0 \times 10^{-6}) = \mathbf{5.0 \times 10^7 \text{ Pa}}$
3. Copper wire – ductile; glass – brittle; rubber band – elastic; gold leaf – malleable; chalk – brittle.
4. Practical activity: the rubber band typically stretches much more before breaking and shows non-linear behaviour at large

extensions, while a thread may obey a linear force-extension relationship over a smaller range before snapping.

#### End of Topic Questions:

1. Ductility: ability to be drawn into a thin wire. Malleability: ability to be hammered/rolled into sheets. Brittleness: tendency to break/shatter with little deformation.
2. Hooke's Law: extension of a spring/wire is directly proportional to the applied force, provided the elastic limit is not exceeded.
3.  $\text{Stress} = F/A$ ;  $\text{Strain} = \Delta L/L_0$ ;  $\text{Young's Modulus } Y = \text{Stress/Strain}$ .
4.  $\text{Strain} = \Delta L/L_0 = 0.0005/1.0 = 5.0 \times 10^{-4}$
5. Steel has much higher tensile strength and stiffness than glass and can withstand large tensile forces without breaking, while glass is brittle and would shatter under similar loads.
6. Elasticity is the ability of a material to return to its original shape after the deforming force is removed; plasticity is the ability to retain the new (deformed) shape permanently.

## Topic 4: Temperature and Thermal Expansion

#### Worksheet 4.1:

1.  $T(K) = 37 + 273 = 310 \text{ K}$
2. The bimetallic strip bends towards the metal with lower expansivity as it heats up (since the other metal expands more); this bending breaks electrical contact, switching off the heating element until it cools and straightens again.
3. Learner experiment: heat equal lengths of different metal rods/wires by the same amount and measure the extension of each to compare linear expansivities.
4. Small gaps allow the concrete to expand on hot days without cracking or buckling due to compressive stress.
5. Learner research - plastics may soften/deform at lower temperatures due to their molecular structure, while glass, though it can withstand higher temperatures, is brittle and cracks under sudden, uneven temperature changes (thermal shock) because different parts expand at different rates.

#### End of Topic Questions:

1. Temperature is a measure of the degree of hotness or coldness of a body (average kinetic energy of its particles).
2. SI unit is the Kelvin (K);  $25^\circ\text{C} = 298 \text{ K}$
3. Any three of: liquid-in-glass thermometer (liquid expansion), bimetallic strip (differential metal expansion), thermocouple (voltage generated at a junction), thermistor/RTD (resistance change), infrared thermometer (detects emitted infrared radiation).
4. Water contracts as it cools from higher temperatures down to  $4^\circ\text{C}$ , but then unusually expands as it cools further from  $4^\circ\text{C}$  to  $0^\circ\text{C}$ , reaching maximum density at  $4^\circ\text{C}$ .
5. Any two of: thermostats in irons/heaters (bimetallic strip), expansion joints on bridges/railways, riveting of metal plates, fire alarms.
6. To allow for expansion on hot days and prevent the tracks from buckling.

## Topic 5: Moments and Equilibrium

#### Worksheet 5.1:

1.  $400 \times 1.5 = 300 \times d \rightarrow d = 600/300 = 2 \text{ m from the pivot}$
2. Learner activity: suspend the cardboard from different points, drop a plumb line from each, mark the lines; the intersection point is the centre of gravity.
3. Learner examples e.g. stable: a table, a cone on its base; unstable: a pencil balanced on its tip; neutral: a ball, a rolling drum.
4. A spanner applies force at a distance from the nut (the pivot); a longer spanner increases the perpendicular distance, increasing the moment (turning effect) for the same applied force, making it easier to loosen the nut.

#### End of Topic Questions:

1. Centre of gravity is the point at which the entire weight of a body appears to act.
2. Stable: returns to original position after small displacement. Unstable: moves further away after small displacement. Neutral: stays in the new position after displacement.
3. Principle of Moments: for a body in equilibrium, sum of clockwise moments about a point = sum of anticlockwise moments about the same point.
4. Torque is the turning effect produced by a couple; SI unit is the newton metre (Nm).

- $2 \times 30 = 4 \times d \rightarrow d = 60/4 = 15 \text{ cm}$  from pivot  $\rightarrow$  mark =  $50 + 15 = \mathbf{65 \text{ cm mark}}$  (on the opposite side from the 20 cm mark)
- Any two of: height of centre of gravity (lower is more stable), width of base (wider is more stable).
- A moment is the turning effect of a single force about a pivot; a couple is a pair of equal, opposite, parallel forces that produce pure rotation.

## Topic 6: Energy, Work, Power and Machines

### Worksheet 6.1:

- $W = F \times d = 150 \times 20 = \mathbf{3000 \text{ J}}$
- $M.A. = \text{Load/Effort} = 250/50 = \mathbf{5}$
- $V.R. = 2/0.5 = \mathbf{4}$ ; Efficiency =  $(M.A./V.R.) \times 100\% = (5/4) \times 100\% = \mathbf{125\%}$  - Note: this unusually high value would suggest a measurement/setup error, since efficiency of a real machine cannot exceed 100%; learners should discuss and identify possible experimental errors.
- Practical activity - learners construct and demonstrate a working lever/pulley system.
- $P.E. = mgh = 2 \times 10 \times 5 = \mathbf{100 \text{ J}}$

### End of Topic Questions:

- Work: force  $\times$  distance moved in direction of force (joule, J). Energy: capacity to do work (joule, J). Power: rate of doing work (watt, W).
- Law of Conservation of Energy: energy cannot be created or destroyed, only transformed from one form to another.
- $M.A. = \text{Load/Effort}$ ;  $V.R. = \text{Distance moved by effort/Distance moved by load}$ ; Efficiency =  $(M.A./V.R.) \times 100\%$ .
- Because some input energy is always lost as heat due to friction between moving parts.
- Any four with examples e.g. Lever - wheelbarrow; Inclined plane - ramp; Pulley - well/flag hoist; Wheel and axle - screwdriver; Gears - bicycle.
- $P = W/t = (F \times d)/t = (500 \times 6)/10 = 3000/10 = \mathbf{300 \text{ W}}$
- At the highest points, energy is mostly gravitational potential energy (maximum height, momentarily at rest); as it swings down, potential energy converts to kinetic energy, which is maximum at the lowest point of the swing.

## Topic 7: Properties of Waves

### Worksheet 7.1:

- $v = f\lambda = 50 \times 4 = \mathbf{200 \text{ m/s}}$
- Learner diagram showing crest, trough, wavelength (distance between two successive crests) and amplitude (maximum displacement from rest position).
- Practical activity: shaking the rope at the correct (resonant) frequency produces nodes (points of no movement) and antinodes (points of maximum movement) along its length.
- As the train approaches, sound waves are compressed, increasing the observed frequency (higher pitch); as it moves away, waves are stretched, decreasing the observed frequency (lower pitch) - this is the Doppler effect.

### End of Topic Questions:

- Transverse: particles vibrate perpendicular to wave direction (e.g. water waves); Longitudinal: particles vibrate parallel to wave direction (e.g. sound waves).
- Wavelength: distance between two successive identical points on a wave. Frequency: number of complete waves passing a point per second. Amplitude: maximum displacement from the rest position.
- $v = f\lambda$
- A travelling wave transfers energy from one place to another; a stationary wave is formed by two identical waves travelling in opposite directions superposing, and does not transfer energy - it has fixed nodes and antinodes.
- Resonance is when an object vibrates at maximum amplitude because the frequency of an external force matches its natural frequency; example: pushing a swing at the right timing to make it go higher.
- Doppler effect: apparent change in frequency of a wave due to relative motion between source and observer; application: speed radar guns, astronomy (red-shift/blue-shift).
- $\lambda = v/f = (3 \times 10^8)/(100 \times 10^6) = \mathbf{3 \text{ m}}$

## Topic 8: Radioactivity and Stability of Isotopes

### Worksheet 8.1:

1. Learner chart comparing alpha (helium nucleus, +2 charge, low penetration, high ionizing power), beta (electron, -1 charge, medium penetration, medium ionizing power), gamma (electromagnetic wave, no charge, high penetration, low ionizing power).
2. 40 days/10 days = 4 half-lives; remaining mass =  $80 \times (\frac{1}{2})^4 = 80/16 = 5 \text{ g}$
3. Although less ionizing per interaction, gamma radiation penetrates deeply into the body and can damage many cells and organs over a large area/depth when a person is exposed to a strong external source, unlike alpha which cannot even penetrate skin.
4. Learner research on carbon-14 dating: comparing remaining C-14 to stable C-12 in a sample to estimate age based on the known half-life of C-14 (about 5,730 years).

### End of Topic Questions:

1. An isotope is any of the different forms of an element with the same protons but different neutrons; a radioisotope is specifically an unstable isotope that emits radiation.
2. Alpha (helium nucleus, low penetration), Beta (fast electron, medium penetration), Gamma (electromagnetic wave, high penetration).
3. Half-life is the time taken for half of the radioactive atoms/activity in a sample to decay.
4. Fission: splitting of a heavy nucleus into smaller nuclei. Fusion: joining of light nuclei to form a heavier nucleus (releases even more energy, powers the sun).
5. Medicine: radiotherapy, diagnostic imaging, sterilisation. Industry: flaw detection, thickness gauging.
6. Any four of: lead shielding, maximum distance from source, minimum exposure time, protective clothing/badges, proper labelled disposal.
7. 12 years/4 years = 3 half-lives; fraction remaining =  $(\frac{1}{2})^3 = 1/8$

## Topic 9: Electrostatics

### Worksheet 9.1:

1. Learner activity: rubbing the pen on hair/cloth charges it, allowing it to attract small pieces of paper due to induced opposite charge on the paper.
2. Contact: touching the charged object directly to the electroscope cap transfers some like charge to the leaf, causing divergence. Induction: bringing a charged object near (not touching) the cap causes charge redistribution; earthing the cap while the charged object is near, then removing the earth and finally the charged object, leaves the electroscope charged oppositely to the inducing object.
3. Sharp points concentrate charge most densely, allowing gradual, safe discharge into the air, reducing the buildup of dangerous voltage that could cause a sudden lightning strike.
4. Learner construction/demonstration activity.

### End of Topic Questions:

1. Like charges repel; unlike charges attract.
2. An object becomes positively charged when it loses electrons; it becomes negatively charged when it gains electrons.
3. Charging by contact involves direct touching to transfer charge; charging by induction involves redistributing charge without direct contact.
4. An electroscope has a metal cap, rod and a thin gold/aluminium leaf in a case; charge on the cap spreads to the leaf and rod, causing the leaf to diverge due to repulsion of like charges.
5. Its sharp point concentrates charge, allowing gradual, safe discharge of atmospheric charge into the ground, preventing sudden dangerous lightning strikes on the building.
6. Removing dust/pollutants from industrial chimney smoke; used in some air purifiers.

## Topic 10: Current Electricity

### Worksheet 10.1:

1.  $R = V/I = 12/2 = 6 \text{ } \Omega$
2. (a)  $R = 2+3+5 = 10 \text{ } \Omega$ ; (b)  $I = V/R = 20/10 = 2 \text{ A}$
3. Learner circuit diagram: ammeter in series with the parallel resistor combination, voltmeter connected across the resistors.
4.  $I = P/V = 2000/240 = 8.33 \text{ A (approx.)}$

5. Learner practical activity confirming a straight-line V-I graph through the origin for an ohmic conductor.

### End of Topic Questions:

1. Ohm's Law: current through a conductor is directly proportional to potential difference across it, at constant temperature.
2. E.m.f. is the total energy per unit charge supplied by a source including that lost internally; terminal p.d. is the energy per unit charge available to the external circuit (less than e.m.f. due to internal resistance).
3.  $E = I(R + r)$
4. Any two of: length, cross-sectional area, type of material, temperature.
5.  $1/R = 1/6 + 1/12 = 2/12 + 1/12 = 3/12 \rightarrow R = 12/3 = 4 \Omega$
6. Electrical power is the rate at which electrical energy is converted/consumed; SI unit is the watt (W).
7. A fuse must be in series so that all the circuit current passes through it; this allows it to break (melt) and cut off current to the whole circuit if current exceeds a safe value. In parallel, it would not carry the full circuit current and would not protect the circuit.

## Topic 11: Introduction to Electronics

### Worksheet 11.1:

1. Conductors: copper wire, aluminium cable. Insulators: rubber glove, glass rod. Semiconductors: silicon chip, germanium diode.
2. Insulators have a very large energy gap between the valence and conduction bands, so electrons cannot gain enough energy at room temperature to jump to the conduction band and conduct current.
3. n-type semiconductors are doped with pentavalent impurities, giving free electrons as majority carriers; p-type semiconductors are doped with trivalent impurities, giving "holes" as majority carriers.
4. Learner research: sunlight striking the p-n junction of a solar cell excites electrons, creating a potential difference (photovoltaic effect) that drives current through an external circuit.

### End of Topic Questions:

1. Conductor: allows current to flow easily. Insulator: does not allow current to flow easily. Semiconductor: conductivity between that of a conductor and insulator.
2. (a) Resistance of a metal conductor increases with temperature; (b) resistance of a semiconductor decreases with temperature.
3. Doping is the deliberate addition of impurity atoms to a pure semiconductor to increase its conductivity by increasing the number of free charge carriers.
4. Intrinsic: pure semiconductor with few free carriers. Extrinsic: doped semiconductor with increased conductivity (n-type or p-type).
5. Semiconductors: diodes, transistors/microchips, solar cells. Superconductors: MRI machines, maglev trains.
6. A superconductor is a material that offers zero electrical resistance; this occurs only below a very low critical temperature (cryogenic cooling required).

## Topic 12: Greenhouse Effect and Climate Change

### Worksheet 12.1:

1. Learner list e.g. burning charcoal/firewood, vehicle emissions, industrial emissions, burning of farm waste, use of fertilisers.
2. Learner explanation: sunlight passes through the atmosphere and warms the Earth; the Earth radiates heat back which is partly trapped by greenhouse gases, keeping the planet warm.
3. Learner poster e.g. planting trees, reducing energy waste, promoting walking/cycling.
4. Learner research on local effects e.g. erratic rainfall, prolonged droughts in ASAL counties, flooding in low-lying areas.

### End of Topic Questions:

1. The greenhouse effect is the natural trapping of heat in the Earth's atmosphere by greenhouse gases, keeping the planet warm enough to support life.
2. Carbon dioxide (fossil fuel burning, deforestation), Methane (livestock, landfills), Nitrous oxide (fertilizers, industry).
3. Global warming is the increase in Earth's average temperature; climate change refers to the broader, long-term shifts in weather patterns resulting from this warming.
4. The ozone layer absorbs harmful ultraviolet radiation from the sun, protecting living organisms from its damaging effects.
5. Any four of: using renewable energy, afforestation, reducing deforestation, improving energy efficiency, promoting public

transport, proper waste management.

6. Deforestation removes trees that would otherwise absorb carbon dioxide, and often involves burning, which releases stored carbon dioxide into the atmosphere, increasing greenhouse gas concentration.

## **Topic 13: Introduction to Space Physics**

### **Worksheet 13.1:**

1. Learner diagram showing Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune in order from the sun.
2. Learner explanation: the universe began from an extremely hot, dense point about 13.8 billion years ago and has been expanding and cooling ever since.
3. Learner research and presentation on a chosen space mission (e.g. Apollo 11, Taifa-1, Mars rovers).
4. Learner activity: the tension in the string provides the centripetal force keeping the ball moving in a circular path, similar to gravity keeping planets in orbit around the sun.

### **End of Topic Questions:**

1. The Big Bang Theory states the universe originated from an extremely hot, dense point about 13.8 billion years ago and has been expanding ever since.
2. Observed expansion of the universe (galaxies moving apart) and cosmic microwave background radiation.
3. Star - produces own light via nuclear fusion; Planet - orbits a star, no own light; Moon - orbits a planet; Asteroid/Comet - small rocky/icy bodies orbiting the sun.
4. Because the sun's gravitational pull is stronger closer to it, requiring planets to orbit faster to maintain a stable orbit, resulting in a shorter orbital period (year).
5. Telescope - used to observe distant celestial objects; Satellite - used for communication, weather monitoring, GPS, remote sensing.
6. Any three of: Astronomer/Astrophysicist, Aerospace Engineer, Astronaut, Satellite Communications Technician, Remote Sensing/GIS Specialist.